

Question

Given the existence of a heteroatom-containing cage-like polyvanadate, where $V : O = 3 : 7$, the content of non-metallic elements is 46.56%, bearing 5 negative charges, but not all V are in their highest oxidation state and the heteroatom is located at the center of the cage, and it is known that all V are in square pyramidal coordination. The descriptions of vertices, edges, and faces in the options refer to polyhedra with vanadium atoms and non-terminal oxygen atoms as vertices.

Which of the following statements is correct:

- A. All other options are incorrect
- B. The heteroatom contained in this acid radical is located in the fourth period of the periodic table.
- C. The heteroatom contained in this acid radical is a short-period element.
- D. The mass fraction of oxygen in this acid radical is 37.8%
- E. The mass fraction of oxygen in the acid radical is 44.8%
- F. The polyvanadate has 64 edges.
- G. The polyoxovanadate has 28 edges.
- H. The polyvanadate has 26 faces.
- I. The polyoxovanadate has 32 faces.
- J. The polyoxovanadate has 18 vertices.

Answer

Correct Answer: **I**

Detailed Explanation

Since the heteroatom is located at the center of the cage, and V has a square pyramidal coordination,

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due to the geometric configuration, the terminal O can only face outwards

, according to Coulombic interactions, it can be inferred that the heteroatom carries a negative charge, that is, a non-metallic ion. (In addition, the mass fraction of the metal element in the vanadium-oxygen framework is $57.71\% > 1 - 46.56\%$, so the heteroatom must be a non-metal)

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According to Coulombic interactions, it can be inferred that the heteroatom carries a negative charge, that is, a non-metallic ion

Therefore, assume the chemical formula of the radical is $(V_3O_7)_xM^{5-}$, at this time, the atomic weight m of the non-metallic element **M** satisfies the equation:

$$\frac{16 \times 7x + t}{(50.94 \times 3 + 16 \times 7)x + t} = 0.4656$$

Substituting $x = 1, 2, 3 \dots$ respectively, it is inferred that when $x = 6$, there is the most reasonable solution $t = 126.9$, which is the I element. Therefore, the heteroatom is I, and options B and C are both incorrect.

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Calculations and reasoning show that the heteroatom is I

Thus, the chemical formula of the radical is $V_{18}O_{42}I^{5-}$, and the calculated mass fraction of oxygen is 39.16%, so options D and E are both incorrect.

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The chemical formula of the radical is $V_{18}O_{42}I^{5-}$

Since each V has a square pyramidal coordination, each V has one terminal oxygen coordination, and the number of vertices is the number of oxygen atoms, 42;

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Number of vertices is 42

The ratio of non-terminal oxygen atoms to vanadium atoms is $(42 - 18) : 18 = 4 : 3$, each V connects four non-terminal oxygen atoms, so the non-terminal oxygen atom is $4 \times \frac{3}{4} = 3$ coordinated; therefore, the number of edges is $(42 - 18) \times 3 = 72$.

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Non-terminal oxygen atom is 3-coordinated

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Number of edges is 72

According to Euler's theorem, the number of faces is $72 - 42 + 2 = 32$, option I is correct, and F, J, H, G are incorrect.

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According to Euler's theorem, the number of faces is $72 - 42 + 2 = 32$